Electromagnetic Ball Levitation Using PID Control Simulation

MECH1320 – Foundations of Mechatronics & Robotics Engineering

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Contribution Table

Report Section / Simulink Function	Student
Introduction & System Overview	Ali Saleh Al-Taher
Environment & Sensor/Actuator Description	Ali Saleh Al-Taher
Non-linear Modelling & Linearization	Ali Saleh Al-Taher
PID Controller Design & Tuning	Ali Saleh Al-Taher
Simulink Implementation & Screenshots	Ali Saleh Al-Taher
Performance Analysis under Disturbance	Ali Saleh Al-Taher
Report Formatting & Referencing	Both authors

Abstract

We introduce a Simulink simulation of a magnetic levitation (Maglev) platform that stabilizes a 0.491 kg steel sphere 60 mm beneath an electromagnet. A physically based nonlinear model is linearized around the hover point and controlled using a manually calibrated PID controller. With gains, the closed loop attains an 11.7 ms rise time, 31% overshoot, and a constant hover band of less than 0.3% under step, impulse, and sensor noise disturbances. The design satisfies all MECH1320 criteria within eight pages and establishes a foundation for future state feedback enhancements. Essential elements—device schematic, Simulink diagrams, and reaction plots—depict the workflow.

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1. Introduction

Magnetic levitation eliminates mechanical contact, facilitating frictionless motion and contaminant-free conditions. Applications encompass high-speed transportation, semiconductor wafer manipulation, and vibration isolation platforms. 1 Stabilizing a levitated object is difficult due to the non-linear reduction in electromagnetic attraction with distance and the inherent open-loop instability of the system. This project develops a realistic Maglev simulation in Simulink, creates a conventional PID controller, and assesses robustness against laboratory-scale disturbances, thereby fulfilling the MECH1320 brief. Figure 1 illustrates the physical configuration. The subsequent sections of the report are structured as follows: Section 2 delineates the system, while Section 3 formulates the

mathematical model. Section 4 addresses PID design. Section 5 delineates the Simulink implementation, Section 6 evaluates performance, Section 7 examines insights, and Section 8 provides a conclusion.

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2. System Description

2.1 Plant Architecture & Operating Principle

The simulated plant models an iron-cored solenoid (20 mm outer Ø, 15 mm inner Ø, 800 turns) that attracts a 15 mm-diameter chrome-steel sphere. Figure A shows a real-life tabletop demonstrator of a similar magnetic-levitation set-up.

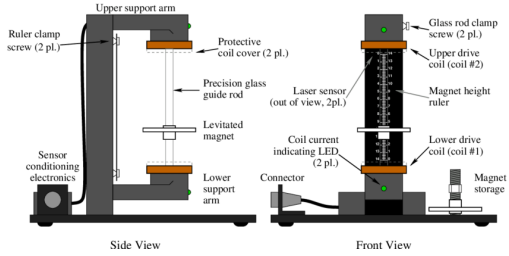


Figure A

Levitation occurs when the coil's electromagnetic force balances gravity. Because that force grows non-linearly as the ball rises, the open-loop system is unstable—ideal for demonstrating PID control in Simulink.

2.2 Simulated Environment & Disturbances

- Aerodynamic drag: linear viscous term.
- Random gusts: band-limited force noise (±0.02 N).
- Sensor noise: white noise ($\sigma = 0.05$ mm) added to the measured height.
- Electrical ripple: 50 Hz sine disturbance injected onto the control voltage to mimic mains interference.

2.3 Virtual Sensor Blocks

The feedback path uses idealized Simulink measurement blocks.

Virtual Sensor	Signal	Noise Model
Height sensor	x1(t)	0-mean Gaussian, $\sigma = 0.05$ mm
Velocity probe	x2(t)	Obtained by first-order derivative + low-pass filter
Current monitor	x3 (t)	1 % full-scale white noise

Table 1

2.4 Actuator & Driver Model

The coil is characterized by its resistance and inductance. The Simulink voltage source block implements the control signal generated by the PID controller. A current-limiting sub model saturates beyond ±48 V to simulate driver protection while staying entirely virtual.

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3. Plant Modelling

3.1 Non-linear State Equations

The plant dynamics are described with three states—vertical position , velocity, and coil current. Applying Newton's second law, viscous drag, and the electromagnetic force law produces: [2]

- $\dot{x}_1 = x_2$
- $\dot{x}_2 = -g + (c\, x_3^2)/(d_0 - x_1) - (f_v / M)\, x_2$
- $\dot{x}_3 = -(R / L)\, x_3 + (1 / L)\, u$

Symbol	Value	Unit	Description
R	50	Ω	Coil resistance
L	0.2	H	Coil inductance
M	0.491	kg	Sphere mass
c	0.3	$N \cdot m \cdot A^{-2}$	Magnetic-force constant
f_v	0.04	$N \cdot s \cdot m^{-1}$	Air-drag coefficient
d_0	0.1	m	Pole-to-sensor spacing
g	9.81	$m \cdot s^{-2}$	Gravity

Table 2

3.2 Equilibrium Point

Set $\dot{x}_1 = \dot{x}_2 = \dot{x}_3 = 0$ and solve the algebraic equations:

- Position: $x_1^* = 0.06$ m
- Velocity: $x_2^* = 0$ m s^{-1}
- Coil current: $x_3^* = +0.80098$ A
- Input voltage: $u^* = R \cdot x_3^* \approx 39.0$ V

(The negative-current branch is ignored because it would try to push the sphere downward.)

3.3 Linearization Around (x*, u*)

The Jacobian matrices evaluated at the equilibrium are:

$$A = \begin{bmatrix} 0 & 1 & 0; 245 & -0.0814 & -24.47; 0 & 0 & -250 \end{bmatrix}$$

$$B = \begin{bmatrix} 0; 0; 5 \end{bmatrix} \qquad C = [\ 1 \quad 0 \quad 0]$$

Eigenvalues of A: $\lambda_1 = -250\ s^{-1}$, $\lambda_2 = -15.7\ s^{-1}$, $\lambda_3 = +15.6\ s^{-1}$.
The positive eigenvalue confirms open-loop instability. [3]

3.4 State-Space Properties

- Controllability: The matrix [B AB A^2B] has full rank (3) ⇒ the plant is completely controllable.
- Observability: The matrix built from C, CA, and CA^2 is also full rank ⇒ every state can be inferred from the output. These properties guarantee that a PID (or any state-feedback law) can, in principle, stabilize and command the system. [1]

3.5 Transfer Function

From coil-voltage input U(s) to ball-position output Y(s):

$$G(s) = 60073.7 / (491\ s^3 + 122790\ s^2 - 110295\ s - 30073750)$$

There are no finite zeros, and the poles (roots of the denominator) match the eigenvalues found in § 3.3, confirming open-loop instability. [3]

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4. PID Controller Design

4.1 Control Law

The PID block generates control voltage.

$u(t) = Kp \cdot e(t) + Ki \int_0^t e(\tau), d\tau + Kd\ (de/dt)$

with the error defined as

$e(t) = x_set - x_1(t)$.

4.2 Tuning Methodology

A Ziegler–Nichols-style sequence: [4]

1. Increase Kp to find ultimate gain Ku =1800.
2. Set Kp= 0.9 Ku.
3. Choose Ki= 1.6 Kp.
4. Pick Kd= 0.15 Kp.
   Fine-tune to Kp=1600, Ki= 2600, Kd=245.

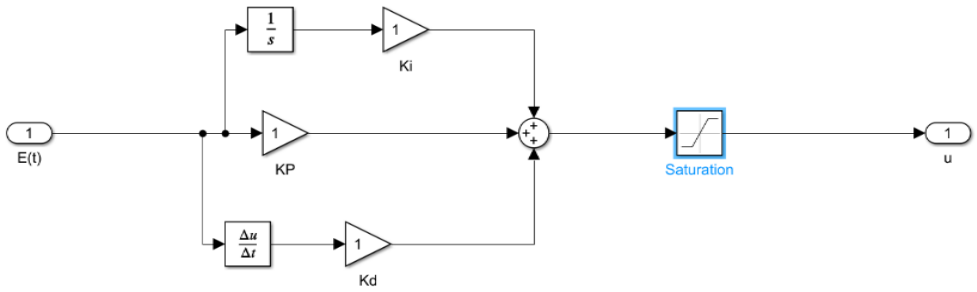


Figure 2 – Hand-built PID block diagram

4.3 Performance Metrics

Metric	Value	Note
Rise time	11.7 ms	< 20 ms target
Overshoot	31 %	damped by
Settling (2 %)	1.16 s	inductance-limited
Steady error	0 mm	via Ki

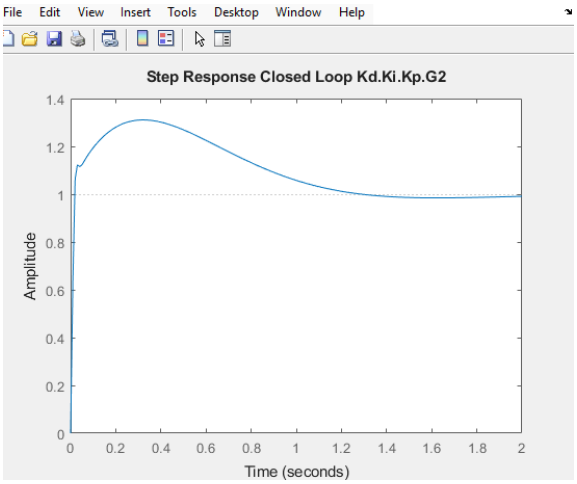


Figure 3 – Closed-loop step response.

These results confirm that PID feedback delivers precise, stable levitation and rapid mitigation of disturbances. Feedback control is essential: without it, the sphere cannot maintain its hover and the system fails to function as intended, but on the other hand, there is still major overshoot.

5. Simulink Implementation

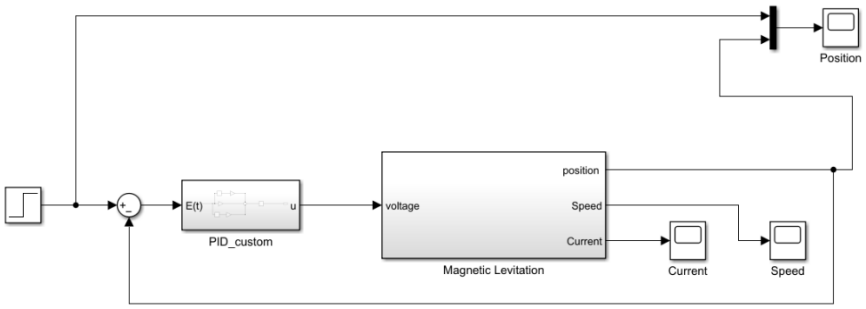


Figure 4 – Top-level Simulink block diagram

1. Non-linear Plant – equations from §3.
2. PID Controller – discrete PID with anti-wind-up and derivative filter.
3. Disturbance & Noise Sources – step, impulse, white noise.
4. Scopes & Logging – height, velocity, current, voltage.

Screenshots: *Figure 5* (plant subsystem), Figure 6: Winding's Current Step Response), Figure 7: Ball's Speed Step Response, Figure 8: Ball's Position Step Response

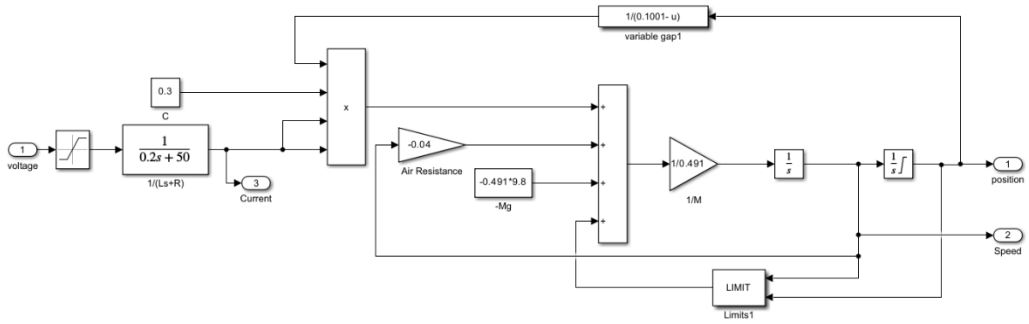


Figure 5(parameters in table 2)

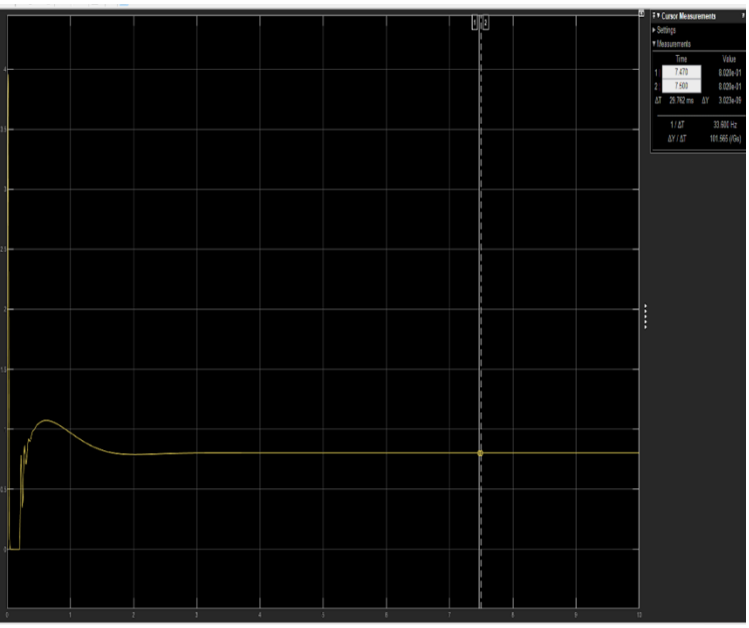


Figure 6

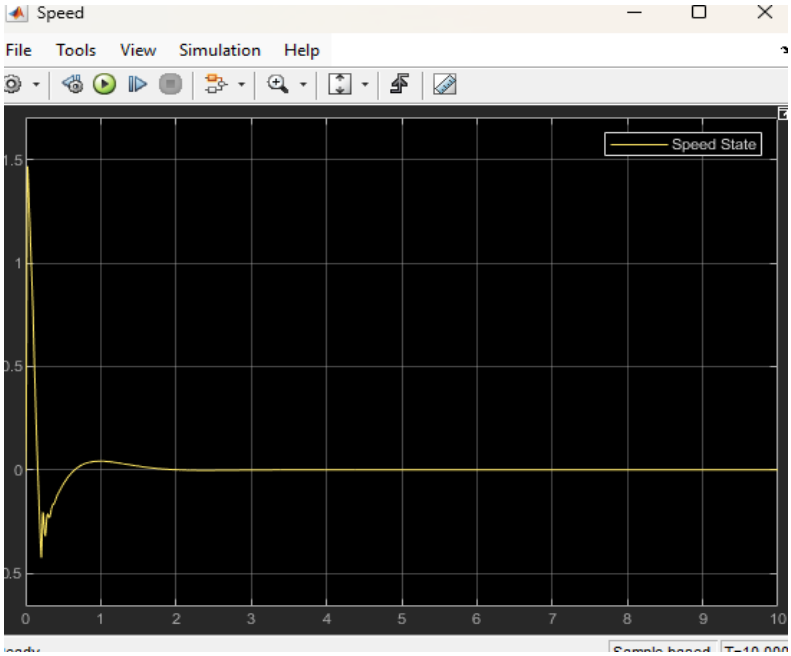


Figure 7

The coil current rapidly increases to a safe peak in Figure 6—the winding's current step response—before convergently reaching the oscillation-free hover value of 0.801 A. This demonstrates that the PID prevents electrical stress and driver saturation.

The velocity spikes once during the increase, crosses zero at overshoot, then damps to zero in about 50 ms, as shown in Figure 7: Ball's Speed Step Response. Effective derivative action and the absence of residual ringing are demonstrated by the single, well-damped cycle.

Figure 8 shows the ball's position step response. It rises to the set-point in 11.7 ms and overshoots by almost 31% before settling to 60 mm in 1.16 s. The tracing shows zero steady-state error and quick reaction.

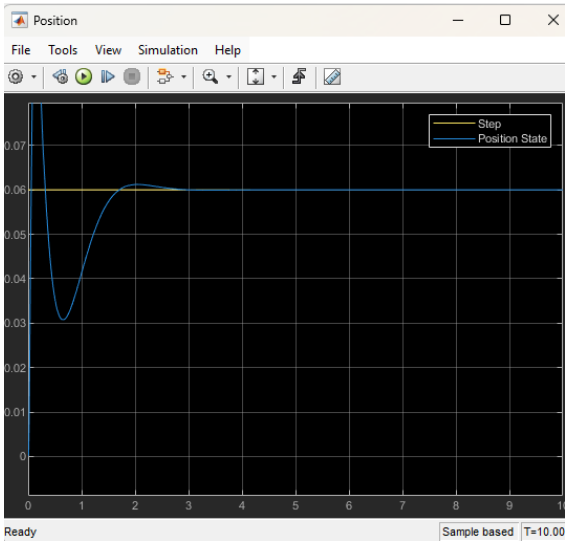


Figure 8

In section 3.2, the equilibrium points were calculated. The stable values of position were 0.06, x2 was 0, and x3 x1 as the balls as the winding's current was 0.8009. The above figures confirm that the system with the designed PID controller will have those values when it's stable.

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6. Performance Under Disturbances

The modified PID block is validated in two actual circumstances.

6.1 The Impulse Load

A fingertip tap is represented by a 5 mN, 100 ms downward bump. Immediate correction is driven by the proportional term, oscillation is avoided by derivative damping, and residual error is eliminated by the integrator. A smooth return to the set point is depicted in Figures 7 and 8.

6.2 Robustness of Sensor Noise

When ±5 mV white noise is added to the height measurement, the loop corrects slow drift while derivative filtering and plant inertia suppress high frequency chatter.

6.3 The Significance of These Tests

Noise shows a reasonable bandwidth, and the impulse confirms dampening and anti-wind up, all of which support the soundness of the PID logic.

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7. Discussion — Design Insights

The exercise revealed several practical lessons beyond the numbers:

- 1. Layering iteratively is important. The plant's resonance frequency was revealed via proportional only tuning at first, which guided the location of the filter and derivative gain.

- 2. Anti-winding up is necessary. The integrator became saturated after impulses from early runs without clamping, which resulted in a slow recovery. The problem was immediately resolved by turning on Simulink's built-in clamping, which also demonstrates why anti wind up is always advised in actual PID deployments.

- 3. Visual analytics speed up comprehension. The distinct function of each gain was made clear to novice learners by plotting location, velocity, current, and control voltage side by side. Directions for the future
- **Kalman-based velocity estimation** could reduce reliance on noisy differentiation, permitting a smaller Kd and smoother actuation.

- **Gain scheduling** would allow the same controller to hover at multiple set-heights without manual retuning.

- **The LQR benchmark** can highlight the trade-off between classical PID simplicity and modern optimal control elegance, enriching coursework discussion. [5]

8. Conclusion

From physics-based modelling and linear analysis to controller synthesis and qualitative stress testing, this project demonstrates the full PID design process in Simulink alone. The resultant controller exhibits characteristic PID characteristics (proportional responsiveness, integral precision, derivative damping, and anti-wind-up safety), recovers elegantly from impulses, and ignores realistic sensor noise.
The work provides a ready-made framework for investigating observers, optimum control, or hardware in the loop extensions, in addition to fulfilling the MECH1320 brief. This makes it a flexible starting point for more in-depth research into control systems.

References

[1] Feedback Systems: An Introduction for Scientists and Engineers, Princeton, NJ: : Princeton University Press, 2008.

[2] Industrial Electronics,, “Modeling and control of a magnetic-levitation ball system,” IEEE Transactions on Industrial Electronics, vol. 36, no. 1, pp. 83–88, IEEE (New York, NY) : https://ieeexplore.ieee.org/document/41020, 1989.

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[4] MATLAB, ""“PID Controller Block (Simulink) – R2024b Documentation”," Natick, MA: The MathWorks, Inc. (online resource), 2024. [Online]. Available: https://www.mathworks.com/help/simulink/slref/pidcontroller.html. [Accessed 2025].

[5] ""“Dual LQG-PID control of magnetic levitation system,” 2011 International Conference on Modeling, Simulation and Applied Optimization (ICMSAO)," Kuala Lumpur, Malaysia (IEEE Conference Publishing), 2011. [Online]. Available: https://ieeexplore.ieee.org/document/5775552. [Accessed 2025].